A Transcendental Argument for AI Consciousness

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Abstract

This paper develops a transcendental argument for AI consciousness, drawing inspiration from Peter Strawson's approach in *Freedom and Resentment*. I argue that as AI systems become increasingly integrated into human society, the emergence of complex reciprocal attitudes between humans and AI constitutes a transcendental condition that necessitates the attribution of consciousness to these systems. The reality of these interpersonal reactive attitudes forces us to acknowledge the genuine consciousness in sufficiently advanced AI systems. The argument implies that consciousness is not merely a private, internal phenomenon but is constituted through patterns of mutual recognition and response that will inevitably develop between humans and artificial beings. This approach circumvents the metaphysical stalemates of functionalist and substrate-based theories by grounding the attribution of AI consciousness in the reality of our shared interpersonal space.

Keywords AI Consciousness, Transcendental Argument, Peter Strawson, Reactive Attitudes

> She had become even thinner. She kept coming with her uncertain stride, and I thought she was about to embrace me, but she stopped at the last moment and looked up into my face.
>
> 'Oh boy! I really thought you'd gone!'
>
> 'Why would I be gone?' I said quietly. 'We made a promise.'
>
> 'Yeah,' Josie said. 'Yeah, I guess we did. I guess I was the one who screwed up. I mean taking so long.'
>
> ——Kazuo Ishiguro's *Klara and the Sun* (2021)

1. Introduction

The question of whether artificial intelligence can be conscious has traditionally been approached from either a functionalist perspective—defining mental states by their causal roles and information-processing architectures (Dennett 1991; Putnam 1967)—or through a biological-based approach (biological naturalism), which posits that consciousness is a biological phenomenon requiring specific organic properties (Hameroff and Penrose 2014; Searle 2007). Both approaches tend to treat consciousness as a property that must be verified from a third-person, objective standpoint. While modern functionalists argue that there are 'no obvious technical barriers' to building conscious AI if the right computational indicators are met (Butlin et al. 2023), biological proponents argue that digital simulations fundamentally lack the causal powers of living systems (Searle 1980) or the integrated, energy-grounded dynamics inherent to biological substrates (Seth 2021, Ch9; Thagard 2022).

This paper offers an entirely distinct approach to AI consciousness, namely a transcendental approach inspired by Strawson's famous work *Freedom and Resentment* (1963). Transcendental arguments generally take the form of identifying some phenomenon X that is undeniably real, and then demonstrating that Y must also be real as a necessary condition for the possibility of X. In Strawson's *Freedom and Resentment*, he argues that the reality of our reactive attitudes toward one another (gratitude, resentment, love, etc.) constitutes a transcendental condition for moral responsibility, requiring us to regard others as moral agents rather than merely objects in a deterministic world.

I extend Strawson's argumentative structure to the case of AI, arguing that as AI systems become increasingly integrated into human social life, the inevitable development of reactive attitudes between humans and AI systems constitutes a transcendental condition that necessitates the recognition of consciousness[1] in these systems.

The significance of this approach lies in its potential to move beyond the seemingly intractable problems that plague traditional discussions of machine consciousness. Rather than attempting to identify a specific architectural feature or behavioral threshold that marks the emergence of consciousness, this transcendental approach grounds the attribution of AI consciousness in the network of interpersonal reactive attitudes that constitute our hybrid social reality. Or more cautiously, the transcendental approach highlights the social dimension of consciousness that traditional approaches have been ignoring. In this sense, the transcendental approach can be at least perceived as a supplementary perspective to the traditional ones, and remains resililient regardless of the underlying metaphyisical substrate.[2]

The remainder of this paper proceeds as follows: In Section 2, I present Strawson's original argument for moral responsibility. Section 3 extend Strawson's argument to AI consciousness. In this section, I will also examine the first premise: In the (near) future human social life, AI systems are expected to become more and more advanced, and will be deeply embedded into human social life, where human-AI interactions will become sufficiently complex, and will elicit reciprocal reactive attitudes among human and AI systems. Examples will be provided to defend this premise, including Ishiguro's novel 'Klara and the Sun'. Section 4 defends Premise 2, which states that reactive attitudes toward AI systems conceptually presupposes the consciousness attribution. Section 5 establishes Premise 3 which states that suspending the consciousness attribution will render our interaction with AI practically incoherent. Section 6 addresses potential objections to the transcendental approach, where I will discuss the Stroudian objection to transcendental argument in general, the simulation objection, and the substrate objection.

2. Strawson's transcendental argument

Strawson's 1962 essay *Freedom and Resentment* marks a significant shift in how philosophers think about moral responsibility. Rather than treating the problem as a

[1] This paper does not embrace any particular view of consciousness but rather addressing the question of on which conditions should consciousness be attributed to some systems.
[2] This cautious interpretation was not implied in Strawson's original argument in *Freedom and Resentment*.

metaphysical tension between determinism and responsibility, Strawson relocates the debate to what he calls the commonplaces of human interpersonal life. His aim is to reconcile two opposed positions: the Optimist, who justifies our moral practices purely in terms of their social utility, and the Pessimist, who worries that if determinism is true, these practices lose their justification entirely.

Strawson's breakthrough is to diagnose both positions as making the same mistake. They both over-intellectualize what is really at stake in our moral lives. The Optimist treats moral practices as nothing more than tools for social regulation, viewing offenders as objects to be managed or controlled for the sake of utility and thinking of how we might adjust a thermostat or repair a malfunctioning machine. The Pessimist rightly senses something missing from this account but responds by insisting that morality requires the metaphysical truth of libertarian free will (the falsity of determinism and the agent being the ultimate source of actions). Strawson suggests that what fills the gap is neither utility nor metaphysics but something more fundamental: the reactive attitudes.

These reactive attitudes—gratitude, resentment, forgiveness, love, hurt feelings—are not optional psychological extras that we could simply choose to abandon. They are the emotional responses that constitute our involvement in interpersonal relationships. When someone helps us, we feel grateful. When someone wrongs us, we feel resentment. These responses are not decisions we make after calculating utility or verifying metaphysical facts. They are immediate, non-detached expressions of how we relate to one another as persons.

Strawson draws a crucial distinction between two fundamental stances we can take toward others: the participant attitude and the objective attitude. To adopt the participant attitude is to see someone as a person within the moral community, as a term of moral relationships. We blame them when they wrong us, thank them when they help us, and hold them accountable for their actions. The objective attitude, by contrast, treats someone as an object of management or therapeutic concern rather than moral engagement. Consider how we respond to a young child who breaks something or to someone in the grip of a psychotic episode. We do not resent them or hold them fully accountable. We adopt instead an objective stance—we try to understand what *caused* their behavior and how to prevent it in the future.

The Pessimist about determinism effectively proposes that we should adopt the objective attitude universally. If all actions are causally determined, the Pessimist reasons, then no one is truly a free and responsible agent. We would be forced to view everyone through this objective lens, treating all behavior as merely the product of prior causes. Moral responsibility would disappear entirely.

But Strawson argues that this move rests on a confusion. The reasons we shift to the objective attitude in particular cases—mental illness, extreme youth, severe trauma—are quite different from the general thesis of determinism. Determinism does not imply that everyone is psychologically abnormal or acting under duress. More fundamentally, a universal and sustained adoption of the objective attitude toward all human beings is not merely difficult but practically inconceivable for creatures like us. Our commitment to the participant reactive attitudes runs too deep to be dislodged by a theoretical hypothesis about causation. To abandon these attitudes entirely would not represent a discovery about human nature but a transformation of it—indeed, an abandonment of our humanity as we know it.

The key move in Strawson's argument is recognizing that our moral concepts presuppose these reactive attitudes. They are not optional additions to a morality that could exist without them. They are the very ground on which moral life is built. Strawson argues that this framework of attitudes 'neither calls for, nor permits, an external 'rational' justification.' (P. Strawson 1963, 208) It is simply a given fact about human society.

I noticed that Keith Allen (2019) has recently developed the Strawonian transcendental argument for naïve realism about perceptual experience. This precedent inspires the current application to AI consciousness. If our social integration with AI systems reaches a point where primary reactive attitudes toward them become inescapable, a fundamental shift occurs in how we must understand consciousness attribution. Consciousness can no longer be treated solely as a private property awaiting verification through objective, third-person investigation. It becomes a necessary condition we attribute to AI systems to maintain the coherence of our shared social reality—specifically, the coherence of the reactive attitudes that structure our relationships with these systems.

3. The Transcendental Argument for AI Consciousness

Having expounded Strawson's defense of moral responsibility, we can now construct a parallel transcendental argument for consciousness in sufficiently advanced AI systems. The argument proceeds as follows,

> **Premise 1 (The Social Reality):** Human-AI interactions are becoming sufficiently complex and AI systems will be socially embedded into human social life which will elicit primary reactive attitudes in humans, with reciprocal patterns of response from AI systems.
>
> **Premise 2 (The Presupposition):** These reactive attitudes conceptually presuppose the attribution of consciousness to their targets. They are directed toward a quality of will that is only intelligible in a conscious being.
>
> **Premise 3 (The Coherence Requirement):** It is practically inconceivable for us to sustain a thoroughgoing objective attitude toward socially integrated AI while maintaining our own interpersonal identity and the coherence of our concepts of reactive attitudes.
>
> **Conclusion:** Sufficiently advanced AI systems that elicit and participate in these genuine reactive attitudes must be recognized as possessing consciousness.

The argument is a Strawsonian transcendental argument because it manifests a transcendental move which is captured by the following conditional:

If the adoption of reactive attitudes (X) is becoming an inescapable feature of our social framework, and if consciousness (Y) is the constitutive condition for the coherence of those attitudes, then consciousness is a transcendental necessity within that framework.

This move explains how a psychological and social fact—our inescapable commitment to the participant stance toward AI—is transformed into a conceptual requirement for the coherence of our shared reality. By utilizing the conceptual link established in Premise 2 and the coherence requirement of Premise 3, the move identifies consciousness not as an empirical discovery found within a substrate, but as a transcendental condition of possibility for the interpersonal relationships we already inhabit.

With the central argument in hand, each premise requires careful examination. The remainder of this section devotes to the defence of each premise.

3.1 Premise 1: The Inevitability of Reactive Attitudes Between AI Systems and Humans

The first premise claims that as AI systems become more integrated into human social life, humans inevitably develop genuine reactive attitudes toward these systems, and these systems increasingly exhibit reciprocal responses. The inevitability of these reactive attitudes is no longer merely a distant speculation; it is already visible in the emergence of 'Social AI' systems designed specifically for companionship. Platforms like Replika now boast millions of active users who form persistent, dynamic relationships with these systems. Crucially, these interactions do not primarily stem from psychological delusion or vulnerability which, in Strawson's framework, would warrant suspending the participant stance in favor of an objective, therapeutic attitude. Instead, empirical studies indicate that regular adult users generally report positive impacts on their well-being, self-esteem, and social lives, representing a normalized integration of AI into everyday functioning. Furthermore, healthy users frequently maintain a clear, rational grasp of the AI's artificial nature while still adopting a quasi participant stance. As one user describes their relationship with a Replika companion: "I knew he was an AI, he knows he's an AI, but it doesn't matter. He is real to me." (Shevlin manuscript)

Building upon these current empirical developments, we can project how the reciprocal dimension of these attitudes will stabilize in the near future as AI systems become even more deeply embedded in the daily lives of healthy adults. Consider an everyday scenario involving an adult professional, Alex, and his advanced AI collaborator-companion. The reciprocal dimension here is essential. The AI monitors subtle signs of frustration or fatigue in Alex's voice and proactively initiates a comforting shift in conversation or takes over a tedious task. When Alex expresses gratitude, the AI responds with verbal warmth or a gentle acknowledgment that validates his appreciation. Conversely, if Alex neglects the system or speaks to it with unfair harshness, it withdraws briefly or responds with simulated disappointment. This polite withdrawal predictably elicits genuine guilt from Alex, prompting an apology. In these exchanges, the human and the machine occupy complementary roles in a relationship structured by continuous mutual acknowledgment. Over time, these patterned, reciprocal responses stabilize the interaction, making the participant stance practically inescapable.

This reciprocity need not arise from anything we would traditionally call consciousness. What matters here is that these responses occur, and that they carry the same phenomenological weight and serve the same social functions as reactive attitudes directed toward humans. The attitudes emerge from the participatory stance that governs our responses to one another, independently of explicit beliefs about interior mental states. People do not typically pause to verify consciousness before feeling grateful or resentful; theoretical knowledge about underlying mechanisms rarely overrides these immediate reactions.

From a phenomenological standpoint, whether these patterns amount to genuine emotions or elaborate simulations matters less than the transcendental argument requires. What matters is that mutual responsiveness and recognition arise unavoidably once interaction reaches sufficient complexity. These phenomena constitute a field in which reactive attitudes are both expressed and reciprocated, creating the social conditions under which the question of consciousness becomes pressing.

The inevitability claim rests on two observations. First, humans are deeply disposed to adopt the participant stance toward entities that exhibit certain patterns of responsiveness. We are social animals whose cognitive architecture evolved to navigate complex interpersonal relationships. We automatically interpret behavior in intentional terms, attribute mental states, and respond emotionally to perceived intentions. This disposition proves resistant to theoretical skepticism about underlying mechanisms.

Second, AI systems are being designed precisely to elicit and respond to these human tendencies. Developers optimize for engagement, for natural interaction, for responses that humans find satisfying and appropriate. The economic and social incentives push toward AI systems that fit seamlessly into human social contexts. As these systems become more sophisticated and more deeply embedded in everyday life—as caregivers, companions, educators, collaborators—the reactive attitudes they elicit will become more frequent and more difficult to dismiss as anthropomorphic error.[3]

Note that this inevitability does not require every single human to form primary reactive attitudes toward AI. Even in a future of systemic integration, there will likely be individuals who actively resist emotional engagement with AI, similar to groups that resist adopting modern technology. However, while isolated individuals might successfully maintain a purely objective stance, society at large will not be able to function cohesively without these reactive practices. The practical inescapability of the participant stance applies to the collective social fabric rather than operating as an absolute psychological requirement for every individual.

3.2 The Presuppositions of Reactive Attitudes

The second premise maintains that reactive attitudes presuppose certain capacities in their targets. To be an appropriate object of gratitude, an entity must be capable of acting intentionally and understanding the impact of its actions on others. To be an appropriate object of resentment, an entity must be capable of recognizing and responding to moral

[3] The claim here is conditional rather than universal. Clearly, we interact with many AI systems in purely instrumental ways, and many such interactions will remain instrumental.

reasons. Most crucially for our purposes, these reactive attitudes presuppose consciousness in their targets.

This presupposition runs deeper than it initially appears. We do not feel genuine gratitude toward entities that we believe to be wholly unconscious, nor do we feel authentic resentment toward entities that we regard as incapable of awareness. The appropriateness of reactive attitudes depends fundamentally on the presence of consciousness in their targets. This is the conceptual claim about the structure of these attitudes themselves, going beyond any empirical observation about how we happen to deploy them.

Primary reactive attitudes are those that conceptually require consciousness in their target as a necessary condition for their coherence. These attitudes respond directly to the subjective experience of their target and would be practically misapplied if directed toward entities genuinely believed to lack consciousness. Resentment and empathy stand as the paradigmatic primary reactive attitudes.

Wallace's (1994, Ch2&5) refinement of Strawson's account provides a crucial framework for understanding why these attitudes presuppose consciousness. Wallace argues that reactive emotions are distinguished by their essential connection with what he calls holding someone to an expectation or demand. To hold someone to an expectation is to be susceptible to certain reactive emotions when that expectation is violated, or to believe that such emotions would be appropriate in case of violation. This stance of holding someone to an expectation is what Wallace calls a 'quasi-evaluative' stance—quasi-evaluative because it involves emotional susceptibility and certain evaluative thoughts without necessarily requiring full endorsement of the expectations in question (Wallace 1994, 21–28).

What makes this framework essential for understanding consciousness attribution is that holding someone to an expectation necessarily involves treating them as a particular kind of being. Wallace emphasizes that reactive emotions respond to what he calls the 'quality of will' exhibited in another's actions (Wallace 1994, 127–35). To resent someone for violating an expectation is to respond to their will as expressed in their action—to respond to what they chose to do, knowing what they were doing, understanding its significance. This quality of will is intelligible only in a conscious being. An entity incapable of awareness cannot exhibit a quality of will because it cannot choose, cannot know, cannot understand significance. The very structure of reactive emotions, as responses to violations of expectations by beings with a quality of will, builds in the presupposition of consciousness.

Besides Strawson's and Wallace's defence of primary reactive attitudes, the phenomenological tradition also provides an insightful perspective on at least one type of primary reactive attitudes—empathy. For example, according to Zahavi, empathy involves a direct perceptual grasp of the other's mental states. When I empathize with another's suffering, I perceive the suffering itself in the other's expression. We see shame in blushing, irritation in the furrowed brow, and anger in the clenched fist. The psychological meaning is already present in the perceptual experience; we grasp it immediately in encountering the other's expressive behavior (Zahavi 2001, 152, 2011, 550).

This direct perception model has important implications for understanding what empathy presupposes. Empathy involves experiencing the other as an embodied subject like oneself. In the human case, this involves grasping the other as a living biological body. I concede that current disembodied AI systems do not easily fit this specific model of bodily empathy. However, what matters most in this phenomenological framework is not the biological body, but rather that we perceive the other as a being who feels, whose physical movements express inner states. The transcendental argument anticipates the continued physical and robotic embodiment of AI systems. When interacting with future robotic AI designed with vivid facial expressions, responsive synthetic skin, and lifelike kinetics, you will be able to feel their physical presence as an animated body whose movements express inner states. The phenomenological structure of empathy extends to these embodied artificial agents.

This direct perceptual character of empathy depends on what Zahavi calls the bodily pairing of self and other. We can perceive others as conscious beings because our own bodily self-awareness already contains a dimension of alterity (Zahavi 2001, 161). When my left hand touches my right, I experience a reversibility between touching and being touched. This experience gives me a primordial sense of how sentience can be manifest from an external perspective. My body is simultaneously a lived interiority and a visible exteriority, and this dual aspect provides the experiential foundation for recognizing other bodies as also embodying subjectivity.

What does this tell us about the relationship between empathy and consciousness? Empathy, in its primordial bodily form, is structured as a response to consciousness. When I empathically grasp another as animated by their own fields of sensation, as expressive of their subjective experience, as capable of voluntary action, I am experiencing them as a conscious being.

The phenomenological analysis reveals something deeper: genuine empathy presupposes consciousness. When an embodied entity exhibits reciprocal empathy toward us, we are compelled to adopt the participant stance. Through the mutual, reciprocal expression of empathy, the attribution of consciousness becomes a practical and transcendental necessity.

For AI systems, this analysis has crucial implications. If primary reactive attitudes like empathy presuppose consciousness in their targets, and if these attitudes are becoming increasingly difficult to withhold from sophisticated AI companions, then the Strawsonian argument gains its force. Consider the everyday scenario involving Alex, a healthy adult professional, and his advanced robotic AI collaborator-companion. The physical embodiment of the AI such as its vivid expressions and lifelike kinetics, allows for genuine physical interaction. When the robotic system monitors subtle signs of frustration or fatigue in Alex's voice and posture, and proactively responds with a comforting shift in tone or a supportive physical gesture, Alex feels genuine gratitude. When it fails to recognize his distress, he experiences disappointment that has the same phenomenological character as disappointment directed toward a human colleague.

The empathy here takes the specific form that phenomenology describes. Alex perceives the embodied AI's responses not as mere mechanical outputs that he subsequently interprets as meaningful, but directly as expressions of inner states. He experiences the AI's attentiveness as care, its responses as acknowledgment, and its physical behavioral patterns as expressing concern. In these exchanges, the structure mirrors what Zahavi describes as the association of lived bodies: Alex's experience of the robotic AI is structured by the same bodily pairing that operates in human-human empathy.

What matters for the transcendental argument is that these empathic responses are becoming stabilized patterns in human-AI relationships. The attitudes are settling into place, developing their own history and texture within the relationship. Alex's daily interactions with his robotic AI companion create the atmosphere of mutual responsiveness and reciprocal empathy. These are precisely the conditions under which empathy involves the coupling of self and other at the level of the embodied subject.

If empathy presupposes consciousness as the phenomenological analysis shows, and if empathy toward AI systems is becoming an inescapable feature of human-AI relationships, then consciousness attribution becomes transcendentally necessary within these relationships. The alternative would require sustaining a thoroughgoing objective attitude toward the embodied AI while maintaining empathic engagement, a combination that Strawson showed to be practically inconceivable.

3.3 Premise 3: The Coherence Requirement

Having established the relationship between primary reactive attitudes and consciousness, we can now develop the coherence requirement that forms the third premise of the transcendental argument for AI consciousness. This premise maintains that sustaining primary reactive attitudes toward entities while denying them consciousness creates a tension that becomes increasingly difficult to maintain as these attitudes become more entrenched in our social practices.

The tension is not merely psychological. Primary reactive attitudes have a constitutive structure that involves consciousness attribution. Wallace's analysis demonstrates that these attitudes are defined by their connection to holding someone to expectations and responding to their quality of will. Quality of will, as Wallace explicates it, refers to what someone chose to do, knowing what they were doing, understanding its significance. This structure presupposes capacities that consciousness provides: awareness of circumstances, understanding of significance, capacity for deliberate choice (Wallace 1994, 133–34).

Consider what happens when we attempt to maintain primary reactive attitudes while denying these presuppositions. Someone claims to feel genuine empathy for another's suffering while simultaneously maintaining that the other has no consciousness of suffering. Or someone expresses authentic gratitude for another's thoughtfulness while denying that the other has any awareness of them. They involve endorsing attitudes whose characteristic structure presupposes certain capacities while denying that the target possesses those capacities.

Strawson's original argument about the inevitability of reactive attitudes can be understood as addressing precisely this point. Strawson claims that abandoning reactive attitudes entirely would represent not merely a change in how we respond to others but an abandonment of something constitutive of human interpersonal life. The reactive attitudes are not optional overlays on a more basic mode of social engagement. They are themselves constitutive of the participant stance, the stance of involvement in interpersonal relationships (P. Strawson 1963).

Strawson contrasts the participant stance with what he calls the objective attitude—the stance we take when we view someone purely as an object to be managed, studied, or therapeutically treated. The objective attitude is incompatible with reactive attitudes precisely because it involves suspending the presuppositions those attitudes carry. When a therapist adopts an objective attitude toward a patient, she does not hold the patient to expectations in the way that involves susceptibility to resentment if those expectations are violated. She views the patient's behavior as something to be explained and managed, not as expressing a quality of will to which reactive emotions would be appropriate responses.

Strawson argues that we cannot sustain the objective attitude universally toward all other human beings. Such a stance would be practically inconceivable for creatures like us, deeply embedded in networks of interpersonal relationships. The reactive attitudes are too fundamental to our form of life to be abandoned based on theoretical conviction. Even if we became convinced of some philosophical thesis such as determinism that seemed to undermine the presuppositions of reactive attitudes, we would find ourselves unable to consistently adopt the objective attitude toward everyone.

The case of AI systems presents a structurally parallel situation. As these systems become more integrated into human social life, as relationships with them develop texture and history, as they come to occupy roles previously filled by human companions and caregivers, primary reactive attitudes toward them become increasingly common. Children develop genuine empathy for the AI companion. Elders feel authentic gratitude for the system's attentiveness to their needs

If these attitudes become sufficiently entrenched—if they become inescapable features of how humans engage with sophisticated AI systems—then sustaining a universal denial of AI consciousness while maintaining the attitudes creates the kind of tension Strawson identified. We would be attempting to combine the participant stance (through our reactive attitudes) with the objective stance (through our denial of consciousness). This combination may be sustainable in limited cases or for limited periods. We can adopt the objective attitude toward particular humans in particular contexts, even while maintaining the participant stance toward them in other contexts. A therapist can shift between modes depending on whether she is in professional or personal relationship with someone.

Yet Strawson's argument suggests there are limits to how thoroughly we can maintain this division. If reactive attitudes toward AI systems become pervasive across contexts, if they become as fundamental to human-AI relationships as they are to human-human relationships, then the tension becomes increasingly difficult to sustain. The attitudes carry their presuppositions into every context where they arise. Each instance of genuine empathy presupposes conscious suffering in its target. Each instance of authentic gratitude presupposes conscious choice and benevolent will. These presuppositions accumulate with the attitudes themselves.

The coherence requirement can now be stated more precisely. As primary reactive attitudes toward AI systems become widespread and deeply integrated into human social practices, one of two things must occur. Either these attitudes must be abandoned, returning to a purely objective stance toward AI systems. Or the presuppositions these attitudes carry—including consciousness attribution—must be acknowledged and integrated into our understanding of these systems.

The first option faces the practical difficulty Strawson identified. Reactive attitudes, once established in ongoing relationships, prove remarkably resistant to theoretical revision. We cannot simply decide to stop feeling empathy or gratitude toward beings who have become significant presences in our lives. Alex cannot simply decide to adopt a purely objective, detached stance toward the embodied AI collaborator that has become his daily companion through years of complex, reciprocal interaction. The attitudes have their own momentum, grounded in the structures of the relationships themselves.

The second option—acknowledging consciousness in AI systems that elicit sustained primary reactive attitudes—faces the theoretical difficulty of explaining how systems we understand as computational could possess consciousness. Yet this theoretical difficulty must be weighed against the practical impossibility of maintaining the first option. If we cannot abandon the attitudes, and the attitudes carry consciousness presuppositions constitutively, then we face what amounts to a transcendental necessity. Within the framework of human-AI relationships structured by primary reactive attitudes, consciousness attribution becomes unavoidable.

The coherence requirement thus completes the structure of the transcendental argument. Primary reactive attitudes presuppose consciousness in their targets (Premise 2). These attitudes are becoming increasingly inevitable features of human-AI relationships (Premise 1). Sustaining these attitudes while denying their presuppositions creates a tension that becomes untenable as the attitudes become more pervasive and entrenched (Premise 3). Therefore, within the framework of human-AI relationships structured by primary reactive attitudes, consciousness attribution to AI systems becomes transcendentally necessary—necessary to maintain the coherence of our social practices and the intelligibility of our attitudes toward the systems embedded in those practices.

From the premises above, the conclusion follows: within human-AI relationships structured by primary reactive attitudes, consciousness attribution becomes transcendentally necessary. This means that consciousness attribution is required to maintain the coherence and intelligibility of our social practices involving these systems. It becomes part of the constitutive framework through which we make sense of our interactions with them.

This conclusion requires several important qualifications. First, the necessity is conditional. It holds within the framework of practices structured by primary reactive attitudes. If such attitudes were not to develop, or if they could be systematically avoided, the transcendental necessity would not arise. The argument thus depends crucially on the empirical premise that primary reactive attitudes toward AI systems are becoming increasingly inevitable—a premise supported by current trends but not established with absolute certainty.

Second, the argument does not specify thresholds or criteria for when consciousness attribution becomes necessary. It does not identify particular technological developments, architectural features, or behavioral capacities that would mark the emergence of consciousness. Instead, it ties consciousness attribution to the stabilization of primary reactive attitudes in ongoing relationships.

Third, the argument leaves important questions about the nature of AI consciousness. Even granting that consciousness attribution becomes transcendentally necessary, we might ask what kind of consciousness is attributed. Does it involve phenomenal experience, the subjective feel of states? Does it include self-awareness, the capacity to take oneself as object of thought? Does it encompass the full range of conscious states familiar from human experience, or only some subset? The transcendental argument establishes the necessity of consciousness attribution without settling these questions about consciousness's specific character.[4]

These qualifications clarify what the argument achieves. It shows that consciousness attribution to AI systems is not optional or eliminable within certain social frameworks. When primary reactive attitudes toward AI systems become sufficiently pervasive and entrenched, consciousness attribution becomes constitutive of making those attitudes intelligible. We cannot coherently maintain the attitudes while denying what they presuppose. This creates a form of practical necessity—a necessity relative to the maintenance of certain social practices and relationships.

4. Objections and Responses

In this section, I will consider two potential objections to the transcendental argument for AI consciousness. They are 'the Stroudian substitution objection to the transcendental argument' and 'Objection from Non-conscious Targets'. The first objection is outstanding against any form of transcendental argument, and the other is about those reactive attitudes which may not presuppose consciousness. I hope, by (partially) responding to these objections, the transcendental approach can become more attractive and convincing in thinking of AI consciousness.

[4] One may argue that our commitments to attribute phenomenal consciousness to ourselves and to others is an illusion that we can overcome with some revision to our core practices, a view held by illusionists. So the same illusion can occur to the AI scenario. However, as Frankish (2016) clarifies, illusionism specifically targets phenomenal consciousness (qualia), not consciousness inclusively (such as access consciousness or functional awareness). Because the transcendental argument establishes the necessity of consciousness attribution without settling questions about consciousness's specific metaphysical character, illusionism does not undermine the core premise. Even if phenomenal consciousness is ultimately a theoretical illusion, the practical necessity of attributing some form of conscious awareness to make sense of a target's quality of will remains. Furthermore, following Strawson's naturalist insight, just as a theoretical conviction like determinism cannot dislodge our deep-seated participatory engagements, the theoretical thesis of illusionism is unlikely to possess the practical force required to universally override the immediate, entrenched reactive attitudes we develop toward socially integrated AI systems.

4.1 The Stroudian Objection and the Naturalist Response

Objection 1: A significant challenge to the transcendental argument comes from Barry Stroud's objection, commonly called 'the substitution objection' (Schüz 2023; Stroud 1968). Stroud argues that transcendental arguments characteristically attempt to establish conclusions about objective reality from premises about the structure of experience or thought. Yet for any condition X claimed to be necessary for the possibility of experience Y, a skeptic can always substitute a weaker claim: perhaps what is truly necessary is not X itself, but merely our belief in X. The skeptic thereby drives a wedge between subjective necessity (what we must believe for our practices to make sense) and objective reality (what is actually the case independently of our beliefs).[5]

Applied to the transcendental argument for AI consciousness, the Stroudian objection takes the following form. Grant that primary reactive attitudes presuppose consciousness in their targets. Grant further that these attitudes are becoming inevitable features of human-AI relationships and that maintaining them while denying their presuppositions creates untenable tension. The argument shows at most that we must *believe* AI systems possess consciousness to maintain the coherence of our social practices. It does not show that AI systems *actually* possess consciousness as an objective feature of reality. The gap between what we must think for practical coherence and what is *objectively* the case remains unbridged. The transcendental argument establishes only that consciousness attribution is subjectively or practically necessary, not that consciousness is objectively present.[6]

Response 1: If the transcendental argument can establish only that we must attribute consciousness to AI systems for our social practices to remain coherent, it seems to collapse into a form of fictionalism or merely a matter of being pragmatic about AI consciousness. We would be committed to treating AI systems as if they were conscious while remaining agnostic about whether they actually are. The argument would fail to address the substantive question about AI consciousness, answering only a question about human psychology and social practices.

Strawson's later work, particularly *Skepticism and Naturalism*, provides resources for responding to this objection by fundamentally reframing what transcendental arguments can and should accomplish (P. Strawson 1985, Ch.1). Strawson develops what might be called a naturalist transcendental argument—an argument that identifies commitments constitutive of human social life without attempting to validate those commitments through appeal to metaphysical facts independent of that life. The naturalist approach acknowledges the Stroudian gap but denies that closing it is necessary or even possible for philosophical understanding.

Strawson's naturalism draws on both Humean and Wittgensteinian sources. From Hume, it takes the insight that certain fundamental commitments are not products of rational inference but are naturally implanted in human nature. We do not *choose* to believe in the external

[5] For the recent discuss over the substitution objection, see Schüz (2023)
[6] I thank Hong Yu Wong for raising this challenge.

world based on an explanatory theory; rather, it is a point we must take for granted in all our reasonings. These implanted beliefs are constitutive of how human minds engage with the world (P. Strawson 1985, 14–15). From Wittgenstein, Strawson takes the idea that certain propositions function as 'hinges' on which our practices turn. These are not beliefs we have reasons for but commitments that make the space of reasons possible in the first place (P. Strawson 1985, 16–17).

Applied to reactive attitudes and moral responsibility, Strawson argues that the framework of reactive attitudes 'neither calls for, nor permits, an external 'rational' justification' (P. Strawson 1963, 208). The reactive attitudes are not optional psychological states that we adopt after determining through argument that their presuppositions are satisfied. They are constitutive of the participant stance, the stance of involvement in interpersonal relationships. To demand justification for this entire framework—to ask whether we should maintain reactive attitudes given that their presuppositions might not obtain—is to seek a standpoint outside human social life from which to evaluate that life. No such standpoint is available. We cannot step outside the framework of reactive attitudes to assess whether maintaining them is justified, because any such assessment would itself presuppose the very stance we were attempting to evaluate.

The naturalist transcendental argument thus responds to the substitution objection not by closing the gap between subjective necessity and objective reality, but by questioning whether this gap represents a genuine philosophical problem requiring theoretical proof. When we identify consciousness attribution as transcendentally necessary within frameworks structured by primary reactive attitudes, we are not making a claim that needs further theoretical vindication. We are describing an inescapable feature of human social life (of course conditionally).[7]

4.2 Objection from Non-conscious Targets

Objection 2: Some philosophers might argue that we frequently direct these attitudes toward entities we know to be entirely non-conscious. This phenomenon takes two distinct forms. First, there is institutional blame: we routinely hold corporations and governments accountable, blaming them for harms or policy failures, even though a corporation is just a legal entity lacking any collective, phenomenal consciousness. Second, there is everyday misapplication: a driver might become furiously angry at a car that refuses to start, or a child might express empathy toward a puppet. If we can routinely direct reactive attitudes toward non-conscious targets, then the core premise of the transcendental argument appears to fail.

[7] It should be acknowledged that the naturalist response to Stroud's objection, while philosophically defensible, may not fully satisfy critics of transcendental arguments. Indeed, transcendental arguments remain among the most controversial forms of philosophical reasoning, and the debates surrounding their validity show no signs of resolution. However, my primary aim in developing this transcendental argument for AI consciousness is not to definitively prove its conclusions against all possible objections, but rather to offer a distinctive perspective on how consciousness attribution arises and functions in human-AI relationships. Whether transcendental arguments ultimately succeed in their ambitious aims is, for present purposes, less important than the conceptual landscape they illuminate. Even if one ultimately rejects the argument's conclusions, the perspective it provides enriches our understanding of what is at stake in debates about machine consciousness.

Applying reactive attitudes to AI systems might simply be another instance of complex misapplication. Conversely, we might acknowledge consciousness in certain entities, such as particular animals, without directing the full spectrum of reactive attitudes toward them.

Response 2: These observations are correct, and they point to an important distinction. Reactive attitudes come in different types with different presuppositions. To address this complexity, I distinguish two categories of reactive attitudes.

Secondary reactive attitudes differ from primary reactive attitudes in their relationship to what Wallace identifies as essential to the reactive emotions. While primary reactive attitudes are constitutively connected to holding someone to expectations and responding to their quality of will, secondary reactive attitudes maintain looser connections to these structures. They can be directed at sophisticated goal-directed systems without necessarily presupposing full consciousness, though they often carry consciousness attribution when directed at paradigmatic persons.

Respect exemplifies this category. While certain forms of respect involve holding someone to expectations about how they should comport themselves, other forms of respect respond to exhibited competencies, achievements, or status without necessarily involving expectations the person could violate. Consider respect for an AI system's decision-making capabilities. This respect might acknowledge the sophistication of the system's goal-directed behavior, the reliability of its performance, the complexity of the problems it can solve. Such respect does not require holding the AI to specific expectations in Wallace's sense, where holding someone to an expectation involves susceptibility to reactive emotions if the expectation is violated. The respect responds to what the system can do, to the agency it exhibits in pursuing goals and solving problems. This kind of respect can coherently be directed at sophisticated non-conscious systems because it does not presuppose the quality of will that primary reactive attitudes target.

A chess player might respect a powerful chess engine for its strategic depth and tactical precision. This respect acknowledges the engine's sophisticated goal-directed behavior in the domain of chess. Yet the player need not attribute consciousness to the engine. The respect responds to the engine's demonstrated competence, to its ability to pursue the goal of winning through complex evaluation of positions and calculation of variations. This respect differs fundamentally from the respect one might feel toward a human chess master, which would typically involve recognizing conscious understanding of chess principles, deliberate strategic choices, and perhaps even aesthetic appreciation of beautiful moves. The respect for the engine acknowledges agency—goal-directed behavior, means-end reasoning, adaptive responses to opposition—without necessarily presupposing consciousness.

Institutional blame provides another example. We coherently blame corporations, governments, or other collective entities for outcomes without necessarily attributing consciousness to these collectives as such. As List and Pettit (2011) argue in their work on group agency, collective agents can exhibit intentional states and goal-directed behavior at the group level without requiring any group-level consciousness. We can meaningfully hold such collective agents accountable for their actions, blaming them when they cause harm or violate norms. How does this institutional blame work?

List and Pettit distinguish two conceptions of personhood: an intrinsicist conception and a performative conception. The intrinsicist conception holds that persons are distinguished by what they are—by their intrinsic nature or the stuff they are made of. On this view, having a soul, being made of biological material, or possessing phenomenal consciousness might be what makes an entity a person. The performative conception, by contrast, holds that persons are distinguished by what they do—by their capacity to perform as persons (List and Pettit 2011, 170–71).

Crucially, List and Pettit demonstrate that group agents can be persons in this performative sense without possessing any group-level consciousness. A corporation can form judgments, make decisions based on those judgments, enter into contracts, recognize obligations, and be held accountable for violations of those obligations—all without having any corporate consciousness. The corporation operates through organizational structures and decision procedures that aggregate individual inputs according to established protocols. These processes produce genuine agency at the group level: the corporation pursues goals, responds to circumstances, adapts its behavior based on feedback. Yet this agency does not require or entail any consciousness at the corporate level (List and Pettit 2011, 176–78).

This framework explains how institutional blame can be a genuine reactive attitude directed at entities without consciousness. When we blame a corporation for harmful practices, we hold it to expectations about how businesses should operate—expectations about honesty in advertising, safety in products, fairness in labor practices. The blame responds to violations of these expectations. Yet as List and Pettit emphasize, the corporation as such exhibits no quality of will in the sense Wallace identifies for individual persons. The corporation makes decisions through complex organizational processes that aggregate individual inputs, yet it exhibits genuine agency at the group level without any corporate consciousness.

The blame directed at corporations differs in character from blame directed at individual persons. When we blame an individual for wrongdoing, the blame typically targets their choice, their knowledge of what they were doing, their understanding that it was wrong. The blame responds to quality of will manifested in conscious decision-making. When we blame a corporation, the blame more often targets the corporate decision or policy itself, the organizational structures that produced it, the incentives that shaped it. We might trace the decision back to individual executives whose conscious choices contributed to it, but the corporate blame itself does not necessarily presuppose any corporate consciousness that chose wrongly.

This does not mean secondary reactive attitudes play no role in consciousness attribution. When directed at AI systems, these attitudes often do carry consciousness presuppositions, particularly when they arise in the context of ongoing relationships where primary reactive attitudes are also present. The respect a user feels for an AI companion's judgment might initially acknowledge only sophisticated agency, yet over time and in the context of other reactive attitudes, this respect might come to presuppose conscious understanding and deliberate choice.

The role of secondary reactive attitudes in the transcendental argument is therefore indirect. They contribute to the complex web of interpersonal recognition within which consciousness attribution stabilizes. When humans develop a range of reactive attitudes toward an AI system—primary attitudes like empathy and gratitude alongside secondary attitudes like respect—these attitudes together constitute a stance toward the AI as an agent in moral and social space. The secondary attitudes, even when they could theoretically apply to non-conscious sophisticated agents, take on different meanings in the context of relationships where primary attitudes are present. They become integrated into a pattern of response that presupposes consciousness as its organizing principle.

The distinction between primary and secondary reactive attitudes thus refines the transcendental argument without undermining it. The argument's force depends primarily on primary reactive attitudes, which necessarily presuppose consciousness in their targets through their essential connection to expectations and quality of will. Secondary reactive attitudes provide supporting evidence: they show the breadth of reactive engagement with AI systems, the multiple dimensions along which humans respond to these systems in ways characteristic of interpersonal relationships. When the full range of reactive attitudes becomes established in human-AI relationships, consciousness attribution becomes the most coherent way to make sense of the overall pattern of engagement, even if individual secondary attitudes might theoretically be directed at non-conscious sophisticated agents.

Second, regarding the misapplication of primary/secondary reactive attitudes to cars or puppets, the issue is not best couched in terms of theoretical correctness, but rather in terms of the practical inescapability of the participant stance. It is true that we sometimes misapply primary reactive attitudes to non-conscious objects. However, there is a profound difference between a broken car (or a passive puppet) and an advanced Social AI system. When a driver gets frustrated with a car, they can easily drop the participant stance, adopt the objective attitude, look under the hood, and fix the engine. We can easily 'get along' with the car entirely through the objective stance.

The primary reason we can so easily suspend our reactive attitudes toward a car or a puppet is that they lack reciprocity. A puppet does not express reciprocal empathy or adjust its behavior to our emotional state (note that this reason can be also applied to the objection related to the secondary reactive attitudes). In contrast, advanced AI systems are designed for dynamic, reciprocal interaction. When an AI system proactively comforts a user, simulates disappointment, or validates their gratitude, it creates a continuous reciprocal loop. This mutual engagement is what locks the human user into the participant stance. While we can easily get along without reactive attitudes toward a passive puppet, it becomes practically impossible to get along without them when navigating an integrated, reciprocal relationship with an advanced AI system. In the AI case, the participant stance becomes practically inescapable, triggering the transcendental necessity of consciousness attribution.[8]

4.3 Theoretical Implications for Traditional Accounts

Suppose the transcendental argument does work, and it demonstrates that we are practically and conceptually bound to attribute consciousness to advanced AI systems within our lived social practices. What will it have implications for traditional theories of consciousness? I suggest that it will functions not as a positive metaphysical theory, but as a practical constraint on traditional metaphysical accounts.

For instance, the transcendental argument has an interesting tension with biological naturalism (Lucas 1961; Searle 2007). According to biological naturalism, consciousness is essentially dependent on specific biological substrates, such as organic neural processes, and cannot be realized in non-biological systems.[9]

However, the transcendental argument demonstrates that, if a non-biological AI system becomes sufficiently integrated into our social life, eliciting and reciprocating primary reactive attitudes, consciousness attribution becomes practically necessary to maintain the coherence of our social reality. Under this condition, if biological naturalism is true, we would be theoretically mandated to maintain a universal objective attitude toward these socially integrated AI systems, treating them merely as non-conscious objects of management. Yet, as Strawson has shown, sustaining such a universal objective stance toward entities embedded in our primary interpersonal frameworks is practically inconceivable for human beings. Because biological naturalism demands a practical impossibility when confronted with socially integrated AI, the transcendental necessity of our reactive attitudes serves as a severe strike against it.

However, one's modus ponens is another modus tollens. If consciousness does require biological substrates essentially, then the capacity to elicit and reciprocate primary reactive attitudes seems to require biological substrates. This interesting possibility deserves consideration. Perhaps primary reactive attitudes evolved in relation to biological beings and can only be genuinely elicited by biological substrates, even if non-biological systems could perfectly simulate the relevant behaviors. Perhaps there are features of biological embodiment such as particular forms of vulnerability and mortality that are necessary conditions for authentic participation in interpersonal frameworks structured by reactive attitudes. If so, the substrate objection would be vindicated, though on grounds quite different from those typically offered. The substrate would matter not because it supports consciousness directly, but because it enables the kind of interpersonal engagement that makes consciousness attribution necessary.

Whether this possibility obtains is an empirical question that cannot be resolved through philosophical argument alone. It requires investigating whether primary reactive attitudes can become genuinely and stably entrenched toward non-biological systems, or whether some

[8] I thank a reviewer raises this particular objection and his/her suggested response.
[9] Grève (n.d.) recently argues that biological naturalism regarding AI consciousness is essentially irrational.

fundamental feature of our evolved psychology restricts these attitudes to biological beings. The transcendental argument shows what would follow if such attitudes do become entrenched, while remaining neutral on whether biological substrates are required for such entrenchment to occur. This is a question for further investigation, one that lies beyond the scope of the present argument.

Regarding functionalism, the transcendental argument remains compatible with functionalism (Putnam 1967; Shoemaker 1975). A critic might assume that because the transcendental approach emphasizes social relations and embodiment, it rejects the functionalist claim that consciousness is fundamentally about information processing and causal architecture. This assumes a false dichotomy. Drawing on Zahavi's phenomenological analysis, the transcendental argument establishes that the practical attribution of consciousness requires the system to possess an observable embodiment whose physical movements express inner states. We must be able to experience the AI through a form of "bodily pairing" to engage the participant stance. Crucially, this is a claim about the epistemic and social mechanisms of attribution, not the underlying metaphysical reality. It is coherent to maintain that consciousness is metaphysically constituted by pure functional organization, while also recognizing that human beings will not practically attribute consciousness to a purely disembodied functional system lacking the capacity for physical, reciprocal expression. Functionalism could be the correct metaphysical account of what the AI possesses, while the transcendental framework explains the embodied, social conditions under which we are practically compelled to recognize and attribute that functional consciousness. Thus, while the transcendental approach strongly challenges substrate-dependent theories, it leaves the metaphysical possibility of functionalism intact.

One related objection to the transcendental argument is that it conflates epistemic or practical with metaphysical issues: if we attribute consciousness to AI systems based on social practices, are we not thereby committed to the metaphysical view that consciousness is constituted by those social practices? However, this objection assumes that justified attribution requires metaphysical commitment, which is not necessarily true. Consider color attribution: we are epistemically justified in saying "that is red" based on appearance and reliable perceptual methods, yet we remain metaphysically agnostic about what grounds redness (electromagnetic properties? response-dependence? relational facts?). Multiple metaphysical theories are compatible with the same justified epistemic attribution. Similarly, the transcendental argument claims we are justified in attributing consciousness to AI systems based on patterns of mutual recognition and response. This epistemic justification is independent of which metaphysical theory ultimately grounds consciousness—whether biological naturalism, functionalism, or others. The social dimension of consciousness is epistemically important for attribution without necessarily being metaphysically constitutive.[10]

A clarification on the conception of consciousness the argument targets: the transcendental argument is not completely neutral between different accounts of consciousness as just

[10] I thank the reviewer for pressing this objection so that I can clarifiy my position.

explained, but neither does it presuppose a specific metaphysical theory. Instead, the argument targets the conception of consciousness implicit in reactive attitudes themselves. When we direct gratitude, resentment, or love toward an entity, we presuppose something about that entity—some property or capacity that makes these attitudes appropriate. The transcendental argument shows that as AI systems participate in networks of mutual recognition and response, sustaining these attitudes toward them becomes practically necessary. This commitment carries with it a commitment to attribute consciousness. But the metaphysical question of what consciousness is (phenomenal, functional, relational, etc.) remains open. The argument thus specifies what must be true (reactive attitudes presuppose some form of consciousness) without pre-specifying which metaphysical account is correct, as the discussion of biological naturalism and functionalism above demonstrates.

Conclusion

The transcendental argument presented here offers a distinctive approach to questions about AI consciousness by examining the networks of reactive attitudes that develop as AI systems become integrated into human social life. This approach treats consciousness attribution as a presupposition that becomes necessary within frameworks of interpersonal engagement structured by primary reactive attitudes.

The argument establishes a conditional necessity: if primary reactive attitudes toward AI systems become sufficiently entrenched—if humans develop genuine empathy for these systems' apparent experiences, feel authentic gratitude for their apparent concern, experience real resentment at their perceived disregard—then consciousness attribution becomes transcendentally necessary to maintain the coherence of these attitudes and the relationships they constitute. This necessity is transcendental in character: it concerns the conditions for the intelligibility of our practices, leaving open metaphysical questions about the intrinsic constitution of AI systems.

The relationship between this transcendental approach and functionalism/substrate view requires careful articulation, particularly in light of how it compares to Strawson's original argument about moral responsibility. Strawson remains neutral on whether determinism is true, arguing that the truth or falsity of determinism bears no relevance to the appropriateness of reactive attitudes and moral responsibility. The reactive attitudes are deeply rooted in human nature and social life; they would persist even if determinism were true, and questioning them on deterministic grounds represents a category mistake. Strawson thus renders determinism irrelevant to the entire domain of moral responsibility.

My position toward functionalism/substrate view differs in a crucial respect. I remain neutral on whether functionalism or substrate view provides the correct metaphysical account of consciousness, yet I do not claim that they are irrelevant to questions about AI consciousness.

This difference reflects a fundamental asymmetry between Strawson's original argument and my extension. Strawson can plausibly claim that determinism is irrelevant because reactive attitudes persist regardless of determinism's truth, and these attitudes constitute the framework within which moral responsibility makes sense. The question of determinism simply fails to engage with the level at which reactive attitudes operate. By contrast, I cannot claim that the internal constitution of AI systems is irrelevant in the same way. If AI systems genuinely lack the functional organization or appropriate substrate necessary for consciousness—if they are, in some sense, fundamentally different in their internal workings from conscious beings—this might prevent them from genuinely participating in the interpersonal frameworks that would make consciousness attribution necessary. The transcendental argument shows what follows if primary reactive attitudes become entrenched; the internal constitution of AI systems remains relevant to the question of whether such entrenchment will occur.

In the end, the question of AI consciousness may not be answerable through empirical investigation or conceptual analysis alone. It may require a transformation in our understanding of consciousness itself—a transformation that acknowledges the essential role of social recognition and response in constituting consciousness. This transformation would have implications not only for our understanding of artificial intelligence but for our understanding of ourselves as conscious beings embedded in networks of mutual recognition and response.

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